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Hybrid Acyclic and Spillback-Aware Scheduling Protocol for Intelligent Traffic Lights Control

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Abstract

Traffic jam is a major issue in modern cities. By increasing delay and pollution, road congestions restrict mobility and are proving to be a menace to public health. Particularly, at closely spaced intersections they are reflected in queue spillbacks between upstream and downstream lanes. In this paper, we address their mitigation as a variation of the well-known capacitated facility location problem. We suggest an integer programming formulation along with a delay-and-queue-based traffic lights control heuristic. Unlike solutions commonly found in the literature, the proposed protocol is simultaneously acyclic, fully decentralized, traffic type independent, and able to handle emergency vehicles. Simulation results show that it outperforms two recent related state-of-the-art schemes in terms of minimising delay and queue length especially in a urban network scenario.

Keywords Traffic light · Scheduling · Optimisation · Spillback

1 Introduction

Nowadays, in most large modern cities getting around has become a real headache for every road user. This situation is always due to the lack of infrastructure capable of meeting an ever-increasing demand. The resulting congestions, have many negative impacts on economy, public health, safety and environment; which is manifested by an increase in delays, stresses, rages, accidents and pollution [1]. Intelligent traffic lights control systems aim to mitigate these problems. They involve leveraging the current state of the traffic at intersections to schedule safe rights of way for vehicles and pedestrians. For decades these technologies have been the subject of massive worldwide research effort [2–4].

Solutions commonly found in the literature are designed based on various requirements and goals [5]. However, decisions are generally made via two parameters namely waiting times (delay) and queue length. Especially in urban networks, when the capacity of a downstream lane is exceeded, upstream queued vehicles remain clogged regardless of the traffic light's colour [6, 7]. Such an event referred to as the queue spillback phenomenon, is detrimental to the performance of any traffic control scheme. Surprisingly, so far only few studies have explicitly tackled this problem [8]. The latter becomes more challenging when traffic is heterogeneous or emergency vehicles' prioritization is needed. Spillbacks can be detected by techniques including Cell Transmission Model (CTM), Link Transmission Model (LTM), Learning-oriented approaches, and Shockwave-based models [9]. Spillback-avoidance solutions involve using offsets calculated via methods that consider vehicle dynamics, perception of vehicles' speed, and the platoon dispersion effect [6, 7].

Furthermore, most studies consider only intersections using a *one-lane-to-one* traffic model, i.e. with lanes in which a single movement is served, whether left-turn, through or right-turn. In the contrast, real-word junctions can contain often lanes that allow different destinations from a single queue. Ignoring such *one-lane-to-many* scenarios can lead to several shortcomings that make traffic control solutions dependent of infrastructure and thus restrain their applicability.

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In this work we aim to design a signal control protocol that is able to quickly adapt to changing traffic patterns by avoiding pre-determined cycles and communication overhead between intersections. Our main contributions are:

- We formulate the adaptive traffic control problem at an isolated intersection as a variation of the well-known capacitated facility location problem;
- We propose for both an isolated intersection and an urban network a dynamic traffic lights control protocol that is acyclic, queue and delay-based, fully distributed and traffic model independent. In addition, our scheme explicitly avoids collisions, mitigates spillbacks and handles vehicles' priority. To the best of our knowledge, such a solution does not exist in the literature;
- We suggest a preemption-free emergency vehicles prioritization strategy to prevent notably deadlock or starvation cases when such vehicles are conflicting [6, 10] ;
- Extensive simulations with various scenarios show that our heuristic-based solution minimises delays and queue lengths especially in spillback prone intersection networks.

The rest of the paper is organized as follows: Section 2 surveys the related contributions; then, the proposed solution is detailed in Section 3 ; the performance evaluation process, the results, and discussions are presented in Section 4 followed by conclusion in Section 5.

2 Related Work

Numerous signal control solutions for both an isolated intersection and an urban network have been developed to prevent road congestion. Irrespective of their assumptions and goals, these works are usually classified into two main categories: the queue-based schemes and the delay-based ones. Backpressure is one of the techniques generally used to model the queuing discipline in lanes. Originally designed for routing in communication networks, the effectiveness of backpressure for traffic signal control has been proved by many studies in recent years [11]. Besides, these pressure-driven schemes, several state-of-the-art contributions resort to strategies that mainly derived from optimisation, fuzzy logic, swarm intelligence or machine learning.

Queue-based schemes include solutions that use the number of vehicles, their cumulative lengths or lane's degree of saturation as input. These schemes have to cope with the short queue delay problem. Indeed, vehicles in lanes with extremely low arrival rates tend to experience excessive delays. Recently, Wu et al. [12] proposed a backpressure routing-based strategy dedicated to an isolated intersection that handles an heterogeneous traffic. However, phases' duration is fixed and lanes are assumed to have an infinite capacity. Li and Jabari [13] proposed a weighted backpressure, i.e. a spillback management scheme that applies higher weights to queues exceeding their lane's capacity. Phase with the highest estimated effective outflow rate receives the priority. However, this strategy uses a small number phases which could be detrimental to network throughput. Noaeem et al. [9] suggested a similar approach but leveraging the shockwave theory to estimate queue length along with parameters such as saturated green time, lanes'effective outflow rate. The resulting scheme is acyclic, considers urban networks and finds the best phase between all possible ones. Unfortunately, the required phase improvement and interphase determination stages are complex and time-consuming. Moreover, the spillback mitigation process does not consider the queue length variations in the downstream lanes.

In the literature, many recent queue-based traffic control schemes also rely on optimisation theory. For instance, Younes and Boukerche [14] proposed to assign a maximum amount of time to each phase in order to guarantee fairness. Even if emergency vehicles prioritization is considered, the suggested scheme is cycle-oriented and time are allotted even to phases with no flow. Gao et al. [15] suggested two metaheuristics, namely harmony search and artificial bee colony to solve the phase scheduling problem. The latter is modelled as a bi-objective optimization problem and is pedestrian-oriented. The resulting solution can be applied to urban networks but from a central controller. Grandinetti et al. [16] presented a Cell Transmission Model-based and network-level scheme that explicitly cope with spillbacks. The phase scheduling problem is considered via a convex program. However, the proposed distributed control protocol requires predetermined cycles. Soon et al. [17] proposed a stigmergy-based stategy. Pheromone intensity and an online epsilon-Support Vector Regression are used to respectively detect the traffic density and forecast the occurrence of congestions; then all the traffic light controllers coordinate themselves to create green waves for vehicles in the identified congested lanes. The latter process regrettably, leads to a communication overhead. Zhang et al. [18] proposed a dual-layer optimization process. The upper layer is dedicated to lane control using reinforcement learning the lower one is for phase control based on prediction of *lane genes*, i.e. lanes' dynamic properties. The intersection which has the least conflicts after expression of all lanes genes should be given priority. This genetic scheme requires to select phases matching the current phase and their successors; unfortunately, this process referred to as chain control is very costly. Liu et al. [19] suggested a similar bi-level optimization scheme that combines macroscopical

description of the traffic status and the microscopical depiction of the vehicle behaviors. Vehicles are randomly rerouted based on a saturation approximation process. This strategy is helpful to reduce time. Regrettably, this solution is cycle-dependent and dedicated to only regular demands. Mohajerpoor and Cai [7] proved conditions for spillbacks occurrence on undersaturated short links. Their proposed optimisation scheme can handle emergency vehicles but is cycle-dependent and is dedicated to only intersections with exactly two phases.

Artificial Intelligence and fuzzy logic are often used with queue-based techniques. Youssef et al. [20] suggested a machine learning-based strategy that uses yearly information to predict the future traffic load on each lane; regrettably, this scheme is intersection shape and real-world data dependent. Kumar et al. [21] combined fuzzy logic and reinforcement learning to explicitly handle priority and emergency vehicles. The proposed solution requires monitoring each vehicle. This could be costly for real-world scenarios.

In the second category one can find solutions that use vehicles' sojourn time as an input. These schemes may increase the average queue length especially if the traffic is bursty and heterogeneous. Among the most recent solutions is DBPC proposed by Wu et al. [12] that combines backpressure with fluid limit techniques. Delay of each queue is estimated according to the head vehicle's waiting time weighted by the flow of vehicles that can be transferred to the downstream lane. Unfortunately, the latter flow is calculated assuming that downstream lanes' capacity is infinite. Oza and Chantem [22] suggested a realtime task scheduling model. In this spillback-aware scheme each phase is considered as a set of non-conficting two lanes. This approach is regrettably, cycle-dependent and uses a fixed number of phases. Oza et al. [23] extended this scheme to urban networks to help recover to spillbacks. The proposed heuristic prioritizes links already experiencing congestions. The number of cycles to use before spillback occurs is calculated among all links with conflicting flows. This predetermined cycles process is very costly and prevents any fully distributed implementation. Zhang et al. [24] proposed a multi-objective optimization model. This strategy is based on a variation of the well-known Non-dominated Sorting Genetic Algorithm (NSGA). However, the proposed scheme does not consider either spillbacks or traffic heterogeneity.

Only few hybrid schemes joining the two approaches above mentioned exist in the literature. In these solutions delay and queue length parameters are combined using weighted factors. For example, Wu et al. [12] proposed linear combination to achieve moderate fairness and low average queue length by choosing appropriate weighting parameters, especially when dealing with bursty or heterogeneous traffic. Unfortunately, this duration of phases are fixed and spillbacks are ignored. Bai and Bai [25] suggested a similar scheme where the non-conflicting movements are defined as maximal weighted independent sets to be found in a lane-based conflict graph. Nevertheless, the resulting strategy is V2I dependent and also does not consider spillbacks. Kumar et al. [26] suggested a scheme where lanes are selected in rounds. Each round contains four selection turns. One lane is selected at each turn without repetition. Then, the amount of time required to empty the queue in that lane is calculated and bounded. However, not only this solution ignores spillbacks, but by selecting just one lane at a time during each phase, it yields low throughputs. Jian et al. [27] used a multi-objective approach that combines improved particle swarm optimisation and genetic-inspired techniques. However, the suggested strategy is cyclic, does not consider spillbacks and assumes uneven arrival rates. These shortcomings could be detrimental to any real-world implementation. Tunc and Soylemez [28] proposed a state feedback control strategy; but the latter is very intersection shape-dependent and requires a fixed number of phases.

3 Proposed Solution

In this section we first discuss the motivations and objectives of this paper. Then we describe the key assumptions before detailing our solution referred to as HATLAS (Hybrid Adaptative Traffic Light Acyclic Scheduling).

3.1 Motivation and Objectives

As shown in Section 2, most existing solutions surprisingly do not consider limiting factors such as downstream lanes' available capacity and queue spillbacks. Moreover, real-world junctions often require traffic to be *one-lane-to-many* like the optional straight/right turn movement suggested by lane 12 in Fig. 1b. In this case, downstream lanes' demand estimation becomes tricky due to upstream vehicles' unpredictable destinations. None of the above-mentioned solutions explicitly consider such intersections. Consequently, to take up all these challenges and be really useful, we believe that an adaptive traffic light control scheme should be acyclic, both queue and delay-based, fully decentralized, spillback-aware, able to handle emergency vehicles priority and traffic heterogeneity. To the best of our knowledge, no solution of this kind exists in the literature. Therefore, we aim to design such an hybrid scheme that avoids collisions while minimising delays and queue lengths.

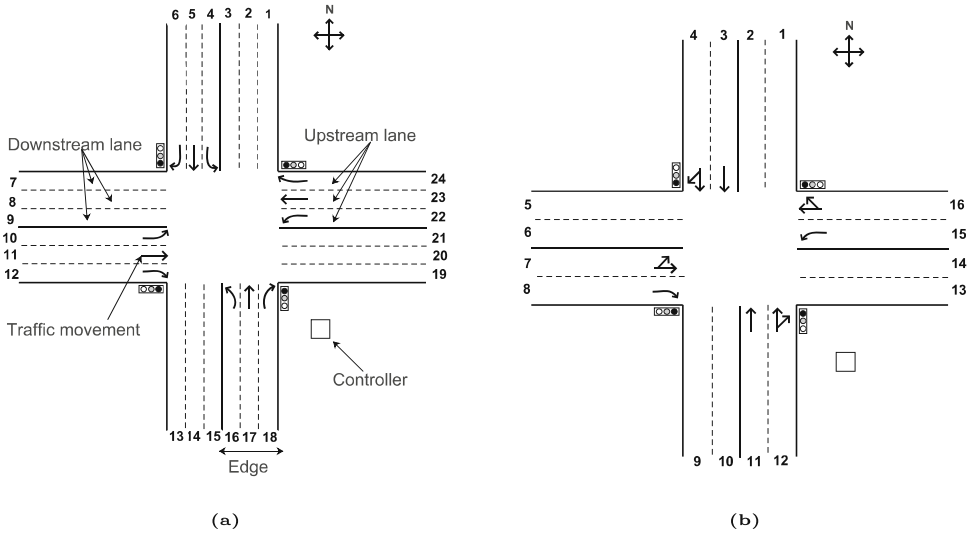


Fig. 1 Typical four-legged intersection with: a) a one-lane-to-one traffic ; b) a one-lane-to-many traffic

3.2 Assumptions

Like most existing studies, we assume that the traffic lights controller is able to:

- get infrastructure information such as Euclidean distance between lanes, the number of lanes, lanes' lengths, traffic movements' types, movements' conflicts;
- estimate real-time traffic state parameters such as vehicles' arrival rates, the number of queued vehicles, vehicles' lengths, inter-vehicle mean distance, lanes' degree of saturation.

We also assume that there is no communication either between vehicles or between controllers.

3.3 Description

We model an isolated intersection as a directed graph $G = (V, E)$ where V denotes the set of lanes and E denotes the set of traffic movements between these lanes. Furthermore, let I and J be two sets that respectively denote the sets of upstream and downstream lanes.$V = I \cup J$ with $|I| = n$ and $|J| = m$; therefore $E = \{(i, j) \in I \times J\}$.

Definition 1 (**Phase**) Let $S^* \subseteq I$ denote the set of upstream lanes i allowed to release a flow of vehicles. A phase is the set denoted by $\mathbf{p}$ such as $\mathbf{p} = \{(i, j) \in E \mid i \in S^*\}$.

Definition 2 (**Edge**) Let ℓ_{ij} denote the Euclidean distance between lanes i and j. The edge of lane i is denoted by the set Γ_i. Formally, $\Gamma_i = \{j \in V | ((i, j) \in I^2) \vee ((i, j) \in J^2) \wedge (\ell_{ij} \approx 0)\}$.

Figure 1a and b depict all the underlying concepts that are commonly found at a typical four-legged isolated intersection. Note that in the remainder of this paper expressions, traffic movement, traffic or movement will be used interchangeably.

The traffic light dynamic control problem lies in leveraging the traffic's current state to define the best rights of way. In other words, find and schedule a phase that is able to avoid collisions and to maximize throughput while minimising vehicles' waiting times. To this end, HATLAS uses a two-stage scheme namely, select the most *urgent phase* to activate and determine its duration.

3.3.1 Most Urgent Phase Selection

Note that unlike most traffic movement-oriented schemes commonly found in the literature, HATLAS uses a lane-based conflict management strategy. This approach aims to find the best phase while considering the *one-lane-to-many* traffic problem as discussed in Section 3.1.

We formulate this traffic light dynamic control problem as a variation of the well-known capacitated facility location problem [29–31]. To this end, we use the following mixed integer linear program:

$$\text{Let} \quad x_i = \begin{cases} 1 \text{ , if any traffic movement from lane } i \text{ is allowed} \\ 0 \text{ , otherwise} \end{cases}$$

Let $y_{ij} \in \mathbb{R}_+$ be the fraction of the capacity (demand) d_j of downstream lane j that is filled when any traffic from upstream lane i is allowed.

Let $a_{ij}=\begin{cases}1\text{ , if at least two movements from lanes } i \text{ and } j \text{ are conflicting}\\ 0\text{ , otherwise}\end{cases}$

$$\min \sum_{i\in I} f_i x_i \tag{1}$$

$$s.t.: \sum_{i\in I} x_i \geq 1 \tag{2}$$

$$\sum_{i\in I} y_{ij} \leq 1,\ \forall j \in J \tag{3}$$

$$\sum_{j\in J} d_j y_{ij} \leq w_i x_i,\ \forall i \in I \tag{4}$$

$$a_{ij}(x_i + x_j) \leq 1\ ,\ \forall i, j \in I \tag{5}$$

$$f_i \in \mathbb{R},\ \forall i \in I \tag{6}$$

Equation 1 states the objective namely, minimize the cummulative fitness score of lanes concerned by each phase. Equation 2 states that at least one lane i must be concerned by each phase. Equation 3 allows a downstream demand not to be entirely filled. Equation 4 ensures that the total supply of a upstream lane i does not exceed its actual capacity w_i. Equation 5 ensures that traffic movements allowed in each phase are not conflicting. Equation 7 defines the fitness score f_i of upstream lane i.

$$f_i = \begin{cases} h_i - c_i^{b_i+1}, & \text{if } c_i \neq 0 \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

$$h_i = \left(1 - \left(\frac{1}{\sigma_1} \times \frac{\psi_i}{\widehat{\psi}}\right)\right) \times \left(1 - \left(\frac{1}{\sigma_2} \times \theta_i\right)\right) \tag{8}$$

HATLAS is aimed to prevent excessive waiting and congestion in each lane. These two events will be indistinctly referred to as a *stroke* in the remainder of this paper.

h_i is the *Stroke index* of lane i given by Eq. 8 where σ_1 , $\sigma_2 \in [0, 1]$ are respectively referred to as *Critical delay ratio* and *Critical saturation ratio*; while ψ_i denotes the sojourn time of the head of vehicles queued in upstream lane i and $\widehat{\psi}$ the *impatience threshold* (i.e. vehicles' maximum bearable sojourn time) provided as a system parameter. c_i and b_i respectively denote the total number of vehicles and the number of emergency vehicles currently in upstream lane i.

$$c_i = \begin{cases} \left\lceil \frac{(\theta_i \times \lambda_i)+\Omega}{(\rho+\Omega)} \right\rceil, & \text{if } \theta_i > 0 \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

c_i is given by Eq. 9; where θ_i and λ_i respectively denote the degree of saturation and length of lane i; with $\theta_i \in [0, 1]$; while Ω and ρ are two system parameters that respectively denote the mean inter-vehicle distance and the length of the shortest vehicle type.

Equation 10 provides T_i the *time-to-next-stroke* in upstream lane i. As for Eq. 11, it helps estimate the *time-to-impatience* denoted by τ_i.

$$T_i = \begin{cases} \min(\tau_i; t_i), & \text{if } c_i \neq 0 \\ +\infty, & \text{otherwise} \end{cases} \tag{10}$$

$$\tau_i = (\widehat{\psi} - \psi_i) \tag{11}$$

Equation 12 gives the *time-to-saturation* of upstream lane i denoted by t_i obtained from c_i, $\widehat{c_i}$ and q_i. These last two parameters respectively denote the maximum capacity of upstream lane i and the vehicle arrival rate. $\widehat{c_i}$ is calculated using Eq. 13.

$$t_i = \frac{(\widehat{c_i} - c_i)}{q_i} \tag{12}$$

$$\widehat{c_i} = \left\lceil \frac{(\lambda_i + \Omega)}{(\rho + \Omega)} \right\rceil \tag{13}$$

Equation 14 helps to assess the demand (i.e. the current capacity) of downstream lane j denoted by d_j.

$$d_j = \begin{cases} \left\lceil \frac{((1-\theta_j)\times\lambda_j)+\Omega}{(\rho+\Omega)} \right\rceil, & \text{if } \theta_j \neq 1 \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

In practice, the value of y_{ij} is not easy to calculate a priori especially when the *one-lane-to-many* traffic model depicted in Fig. 1b is applied. Indeed, traffic controller has no means to predict the destination lane of each vehicle. For example, as shown in Fig. 2 controller cannot determine the fraction of vehicles queued in upstream lane 12 that will meet the demands of downstream lanes 1 and 13. Moreover, it should be also considered that traffic from an upstream lane could overflow on the lanes located on the edge of the destination lane. For instance, as also depicted in Fig. 2 some vehicles from lane 12 are trying to move towards lane 2 since lane 1, their initial destination, is getting clogged. Therefore, to calculate w_i it should be considered to transform y_{ij} into a binary parameter and to relax Eq. 4 with Eq. 15. As well, constraint suggested by Eq. 3 must be abandoned.

So let $y_{ij}=\begin{cases}1\text{ , if movement from lane } i \text{ to lane } j \text{ is possible}\\ 0\text{ , otherwise}\end{cases}$

$$d_j y_{ij} \geq w_i x_i,\ \forall i \in I, \forall j \in J \tag{15}$$

Let $e_{ij}=\begin{cases}1\text{ , if lanes } i \text{ and } j \text{ are on the same edge}\\ 0\text{ , otherwise}\end{cases}$

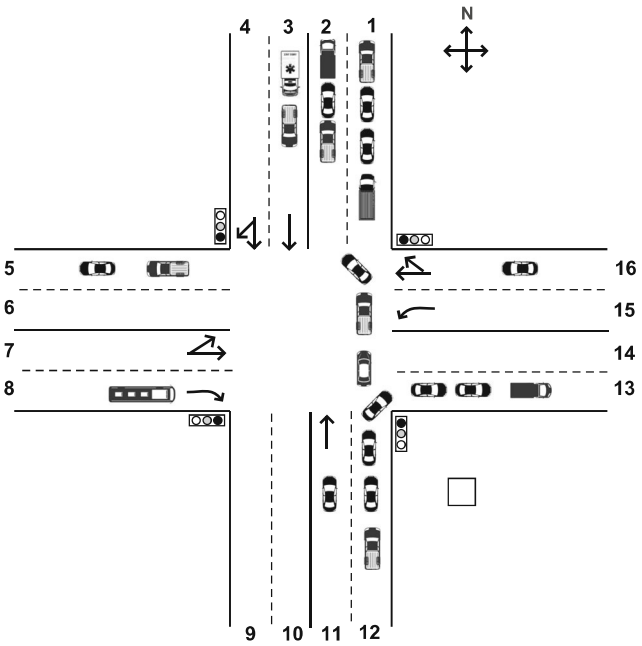


Fig. 2 Edge overflow due to drivers' lane changing behaviour to face an early spillback at an intersection applying the one-lane-to-many traffic model

Equation 16 helps estimate flow ϕ_{ij} that denotes the number of vehicles allowed to move from upstream lane i to downstream lane j considering the possible traffic owerflow on the edge of lane j; where $\xi(i, j)$ denotes the *Spillover effect* of flow ϕ_{ij}, i.e. the number of vehicles from upstream lane i that can move to the neighbouring lanes of the downstream lane j.

$$\phi_{ij} = \begin{cases} d_j + \xi(i,j), & \text{if } y_{ij} = 1 \\ +\infty, & \text{otherwise} \end{cases} \tag{16}$$

$$\xi(i,j) = \sum_{k \in J} d_k \times e_{jk} \times [y_{ik} = 0] \times [\mu(i,k) = 0] \tag{17}$$

$$\mu(i,k) = \sum_{(l \neq i) \in I} x_l \times y_{lk} \times c_l \tag{18}$$

Note that in Eq. 17 the [S] expression denotes the Iverson bracket for statement S such as [S]=1 if S is true and 0 otherwise. Equation 18 estimates $\mu(i, k)$, i.e. the number of vehicles from any upstream lane that could conflict those from upstream lane i when reaching downstream lane k. Equation 19 helps to derive w_i the actual number of vehicles to be allowed from upstream lane i especially if a *one-lane-to-many* scheme is applied (see Fig. 2).

$$w_i = \min(c_i, \phi_{i1}, ..., \phi_{in}) \tag{19}$$

Moreover, to satisfy the constraint suggested by Eq. 5, a method commonly used in the literature is to construct a conflict graph and find all its MISs (Maximal Independent Sets). As claimed above, to avoid collisions while considering the *one-lane-to-many* problem (see Section 3.1), HATLAS creates an undirected graph of which vertices refer to the upstream lanes and edges denote the conflicts between them. Two lanes are said to be conflicting if at least two incompatible movements from them exist. Figure 3a depicts the crossing and converging conflicts at a typical four-legged junction applying both *one-lane-to-one* and *one-lane-to-many* traffic models. The resulting lane-based conflict graph is shown in Fig. 3b. For example in this graph, vertex 12 is respectively linked to vertices 7,11,15 and 16 since simultaneous traffics from the corresponding lanes could lead to collisions. Therefore, finding the most urgent phase requires to list all MISs of such an upstream lane-based conflict graph. Regrettably, this graph theory

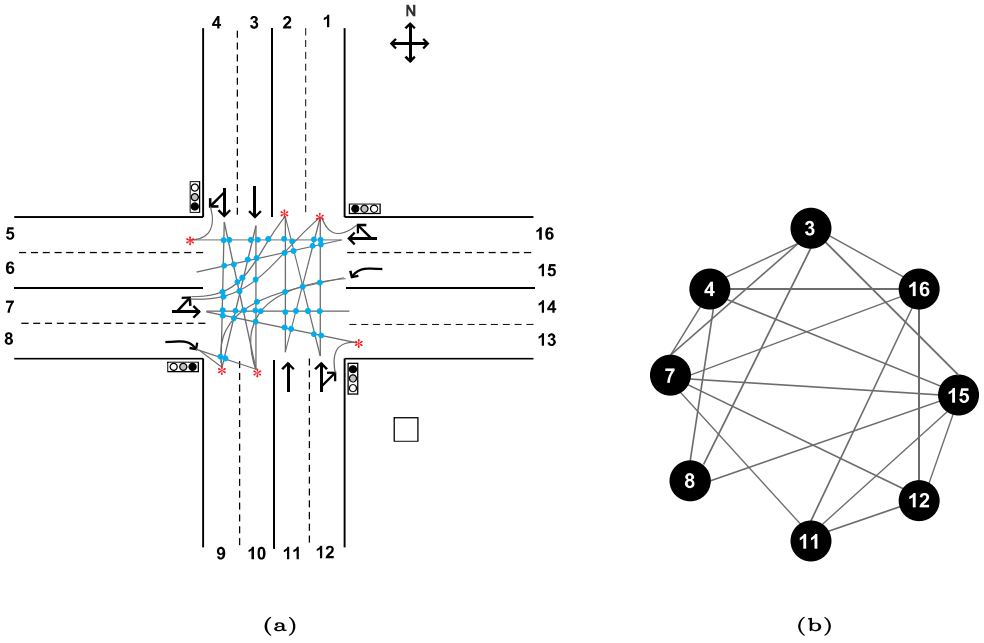


Fig. 3 Conflicts in an isolated intersection: a) points are for crossing conflicts and stars for the converging ones b) corresponding upstream lanes conflict graph

well-known problem has been proven NP-hard [32]. Mooon and Moser [33] showed that any graph with n vertices has at most $3^{n/3}$ cliques hence MISs. Lawler et al. [32] proved that some polynomial time algorithms can be used for special cases. Many quite efficient state-of-the-art solutions can be used in our situation [34–36].

Additionally, we also have to cope with the inherent NP-hardness of facility location problems [37, 38]. Consequently, we propose the following heuristic:

- step 1 find all the MISs from the constructed upstream lane conflict graph;
- step 2 using Eqs. 7–9 calculate the total score of each MIS from the fitness score of each lane it contains;
- step 3 choose the MIS with the lowest total score as the selected phase;
- step 4 using Eqs. 14–19 assess for each selected lane the number of vehicles to allow;
- step 5 calculate the selected phase's duration and try to find a possible more urgent phase.

3.3.2 Duration of the Selected Phase

After selecting the most urgent phase to activate, traffic controller has to calculate its optimal duration denoted by $\widehat{\delta}$ using Eq. 20. δ_{ij} denotes the duration of movement between lanes i and j estimated with Eq. 21 where ℓ_{ij} denotes the Euclidean distance between lanes i and j while $\overline{v}_{ij}$ denotes the average speed to travel the latter distance.

$$\widehat{\delta} = \max(\delta_{11}, ..., \delta_{nm}) \tag{20}$$

$$\delta_{ij} = \begin{cases} \frac{(w_i(\rho+\Omega)-\Omega)+\ell_{ij}}{\overline{v}_{ij}} \times x_i \times y_{ij}, & \text{if } w_i > 0 \\ 0, & \text{otherwise} \end{cases} \tag{21}$$

To maximize the throughput the proposed heuristic favours MISs with largest cardinalities. Nevertheless, it is also helpful to investigate the singletons, i.e. lanes that are not involved in the selected phase in order to find a possible more urgent one. To do this, using Eq. 22 controller looks for the lowest *time-to-next-stroke* of upstream lanes not concerned by the selected phase denoted by $\widehat{T}$; the latter is compared to $\widehat{\delta}$ the duration of the urgent phase selected so far. Hence, if $\widehat{T}$ is lower than $\widehat{\delta}$ or if their difference is lower than $2\tilde{t}$, $\widehat{\delta}$ is updated according to conditions defined by Eq. 23; where $\tilde{t}$ denotes the yellow light duration defined as a system parameter.

Figure 4a–b depict the 3 possible scenarios that require an update of $\widehat{\delta}$. However, the notable difference is that if the event suggested by Fig. 4b occurs, then a new phase is also selected. The latter will only include movements of the lane with the lowest *time-to-next-stroke* that has been found.

$$\widehat{T} = \min_{i \in I \setminus S^*} T_i \tag{22}$$

$$\widehat{\delta} = \begin{cases} 2\tilde{t}, & \text{if } \widehat{T} - 2\tilde{t} \leq 0 \\ \widehat{T} - 2\tilde{t}, & \text{otherwise} \end{cases} \tag{23}$$

Table 1 describes the parameters used by HATLAS. Algorithm 1 details the urgent phase selection and duration assessment process used by HATLAS.

Input: $I, J, ...$ ▷ see Table 1
Output: selected phase $\mathbf{p}$ and its duration $\widehat{\delta}$

$\widehat{\delta} \leftarrow 0$
$\mathcal{L}_{MIS} \leftarrow$ get all the MISs of $\mathcal{A}$
for each set $S \in \mathcal{L}_{MIS}$ **do**
  $r \leftarrow 0$
  $r^* \leftarrow +\infty$
  for each lane $i \in S$ **do**
    calculate h_i, c_i and f_i ▷ see Eqs. 7–9
    $r \leftarrow r + f_i$
  end for
  if $r < r^*$ **then**
    $r^* \leftarrow r$
    $S^* \leftarrow S$
  end if
end for
for each lane $i \in I$ **do**
  $T_i \leftarrow +\infty$
  if $(c_i \neq 0)$ **then**
    calculate τ_i and t_i ▷ see Eqs. 11–13
    $T_i \leftarrow \min(\tau_i; t_i)$
  end if
  for each lane $j \in J$ **do**
    calculate δ_{ij} ▷ see Eqs. 14–21
    if $(\delta_{ij} > \widehat{\delta})$ **then**
      $\widehat{\delta} \leftarrow \delta_{ij}$
    end if
    if $(i \in S^*) \wedge ((i, j) \in E)$ **then**
      $\mathbf{p} \leftarrow \mathbf{p} \cup \{(i, j)\}$
    end if
  end for
end for
$\widehat{T} \leftarrow \min_{i \in I \setminus S^*} T_i$
$\widehat{i} \leftarrow \arg\min_{i \in I \setminus S^*} T_i$
if $(\widehat{T} < \widehat{\delta}) \vee (\widehat{T} - \widehat{\delta} < 2\tilde{t})$ **then**
  if $(\widehat{T} - 2\tilde{t} \leq 0)$ **then**
    $\mathbf{p} \leftarrow \{\widehat{e} \in E : \widehat{i} \in \widehat{e}\}$
    $\widehat{\delta} \leftarrow 2\tilde{t}$
  else
    $\widehat{\delta} \leftarrow \widehat{T} - 2\tilde{t}$
  end if
end if

Algorithm 1 HATLAS' phase selection and duration assessment process.

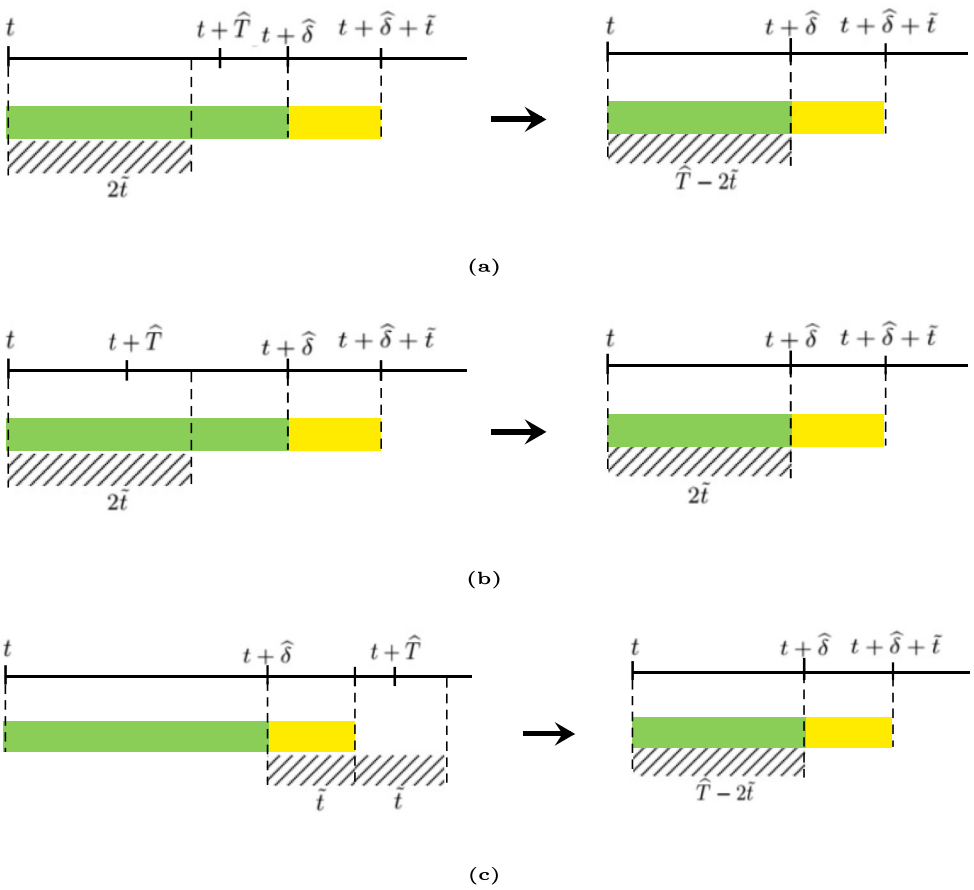


Fig. 4 Possible scenarios for new a phase selection or duration update: a) when $(\widehat{T} < \widehat{\delta}) \wedge (\widehat{T} - 2\tilde{t} > 0)$ b) when $(\widehat{T} < \widehat{\delta}) \wedge (\widehat{T} - 2\tilde{t} \leq 0)$ c) when $(\widehat{T} > \widehat{\delta}) \wedge (\widehat{T} - \widehat{\delta} < 2\tilde{t})$

4 Performance Evaluation

To evaluate HATLAS we conducted a simulation campaign via two scenarios respectively with an isolated intersection and a grid-based network of intersections. Traffic demands were generated with SUMO 1.12.0 [39] linked to OMNeT++ 5.6 [40]. The latter was used to implement all the evaluated solutions. The performance was analyzed through three metrics namely Average delay, Average priority delay, and Degree of saturation. Results are compared with those obtained with WBPC by Wu et al. [12] and DESRA by Noaeen et al. [9] two state-of-the-art protocols that are respectively, hybrid and acyclic as well as spillback-aware. Tables 2, 3 and 4 show the simulation parameters. For the isolated intersection depicted in Fig. 5a, average arrival rates per upstream edge were scaled with parameter α. For homogeneous traffics a 450 veh/h average arrival rate were set for all the lanes; for the heterogeneous ones, 450 veh/h and 900 veh/h average arrival rates were respectively used for lanes on North/South and West/East edges. α were increased from 1 to 2 with a 0.2 step. Figure 5b shows the 5 × 5 grid network of intersections used for urban scenarios. Note that we used a 5 s slot length (T_s) and 0.5 for each weight parameter for WBPC.

4.1 Average Delay

We studied the average delay under different traffic demands at both the isolated intersection and the urban network to evaluate the three protocols' ability to reduce waiting times. Therefore, queued vehicles' waiting times were assessed on each upstream lane and averaged every 1.5 s. Each experiment was repeated 35 times. Results are averaged with a 95% confidence interval.

Figure 6a depicts the results obtained when homogeneous traffic was considered at the isolated intersection; we observe that HATLAS provides the lowest delays irrespective of the traffic intensity. This is due to the impatience control scheme used by HATLAS for its delay-based strategy. Indeed, unlike WBPC, HATLAS uses drivers' impatience threshold to prevent vehicles to exceed the defined maximum sojourn time independently of queue length. DESRA yields the worst values due to its strict queue-based strategy. Figure 6b shows that under heterogeneous traffic results follow the same pattern. However, WBPC has an irregular performance due to its parameter denoted by departing vehicles rate which values can vary.

Figure 7a and b show the results obtained with the urban network scenario. In such situations that are prone

Table 1 Summary of notations

Symbol	Definition
$\mathcal{A}$	Matrix of conflicts between upstream lanes
a_{ij}	Relationship between upstream lanes i and j
b_i	Number of priority vehicles in upstream lane i
c_i	Number of vehicles currently in lane i
$\widehat{c_i}$	Capacity (i.e. maximum number of vehicles) of lane i
d_j	Demand (i.e. current reception capacity) of downstream lane j
δ_{ij}	Duration of queue clearance from lane i to lane j
$\widehat{\delta}$	Duration of the selected phase
E	Set of possible movements between lanes
e_{ij}	Edge relationship between lanes i and j
f_i	Fitness score of upstream lane i
Γ_i	Edge of lane i
I	Set of upstream lanes
J	Set of downstream lanes
ϕ_{ij}	Allowed flow from lane i to lane j
λ_i	Length of lane i
ℓ_{ij}	Euclidean distance between lanes i and j
m	Cardinality of I
$\mu(i,k)$	Number of vehicles from any upstream lane that could be in conflict with those from upstream lane i when reaching downstream lane k
n	Cardinality of J
Ω	Inter-vehicle mean distance
$\mathbf{p}$	Selected phase
q_i	Vehicle arrival rate in upstream lane i
$\widehat{\psi}$	Driver impatience threshold
ψ_i	Head-of-queue's waiting time in lane i
ρ	Length of the smallest vehicle type
σ_1	Critical delay ratio
σ_2	Critical saturation ratio
t_i	Time-to-impatience in lane i
τ_i	Time-to-saturation of lane i
T_i	time-to-next-stroke in upstream lane i
$\tilde{t}$	Yellow light duration
θ_i	Degree of saturation of lane i
$\overline{v}_{ij}$	Vehicle mean speed from lane i to lane j
w_i	Actual number of vehicles to allow from upstream lane i
x_i	Traffic allowance from upstream lane i
$\xi(i,j)$	Number of vehicles from upstream lane i that can spread on the edge of downstream lane j
y_{ij}	The fraction of downstream lane j capacity that is filled when traffic from upstream lane i is allowed

Table 2 Parameters for the isolated intersection

Parameter	Value
Lane length	135 m
Number of lanes	3 per edge
Homogenous traffic	450 veh/h per upstream edge
Heterogenous traffic	(N/S) 450 (W/E) 900 veh/h
Yellow state duration	1.5 s
α (Scale)	1–2
Ω	2.5 m
$\widehat{\psi}$	300 s
ρ	2.2 m
σ_1	0.8
σ_2	0.2
Experiment duration	10000 s
Warm-up	100 s

Table 3 Parameters for the vehicle types

Name	Length	Probability	Priority
Personnal	5 m	20%	
Bus	12.5 m	2%	
Truck	16.25 m	1%	
Motorcycle	2.2 m	1.8%	
Ambulance	8 m	0.2%	✓
Police	8 m	0.5%	✓
Firebrigade	8 m	0.1%	✓

Table 4 Parameters for the urban network

Parameter	Value
Model	Grid 5×5
Lane length	200 m
Number of lanes	3 per edge
Number of vehicles	1000–2500
Arrival distribution	Poisson ($q_i = 3$ veh/s)
Mobility model	Mahattan(25% right, 50% straight, 25% left)
$\tilde{t}$	1.5 s
Ω	2.5 m
$\widehat{\psi}$	300 s
ρ	2.2 m
σ_1	0.8
σ_2	0.2
Experiment duration	12000 s
Warm-up	600 s

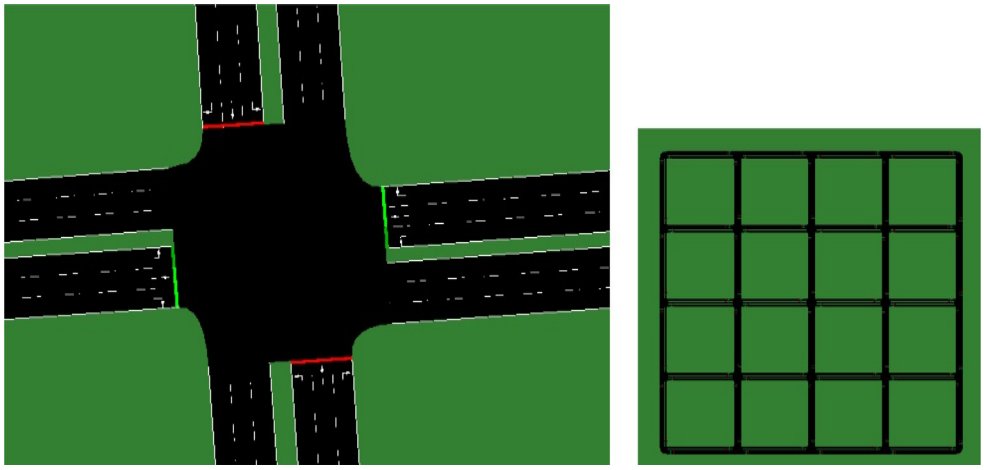


Fig. 5 Implementation of the two simulation scenarios in SUMO: a) four-legged and 3 lanes per edge isolated intersection b) 5 × 5 grid network of intersections with Manhattan mobility

to spillbacks, one can observe that delays increase with the number of vehicles. However, HATLAS provides the lowest delays even under heterogeneous traffics. This is due to the edge overflow control used to cope with spillbacks. Indeed, this scheme tends to maximize the number of departing vehicles during each phase. In contrast, despite its spillback control scheme, DESRA as a strict queue-based solution, struggles to reduce delays compared to WBPC and HATLAS due to their hybrid approach.

4.2 Average Priority Delay

These experiments aimed to assess the ability of the three protocols to specifically reduce emergency vehicles' sojourn times. Simulations were conducted under an heterogeneous traffic at both the isolated intersection and the urban network. Queued emergency vehicles' waiting times were assessed on each upstream lane and averaged every 1.5 s. Each experiment was repeated 35 times. Results are averaged with a 95% confidence interval.

Figure 8b shows the results obtained with the isolated intersection. Once again, HATLAS provides the lowest delays regardless of the traffic intensity. This is due to the priority control scheme that HATLAS uses during its most urgent phase selection process. Indeed, unlike WBPC, HATLAS can explicitly manage emergency vehicles' rights of way while defining lanes'scores without preempting the current phase. DESRA has the worst values due to its fully queue-based strategy.

Figure 9a shows that for the urban network scenario, when using HATLAS delays remain low. Unlike WBPC and DESRA, HATLAS alleviates the time-consuming deadlock and starvation cases by breaking ties just using impatience ratio, degree of saturation, and the number of emergency vehicles when they get stuck in conflicting lanes.

4.3 Degree of Saturation

In this study, we aimed to investigate the three protocols' capacity to prevent lanes' congestion. Queue and lane lengths ratios were calculated for each upstream lane and averaged every 1.5 s. Each experiment was repeated 35 times. Results are averaged with a 95% confidence interval.

Figure 9a suggests that for the isolated intersection with homogeneous traffic scenario, hybrid solutions (HATLAS and WBPC) yield the lowest degrees, around 57%. However, the edge overflow control applied in its spillback mitigation strategy helps HATLAS to further reduce queue lengths. In contrast, DESRA provides the highest degrees

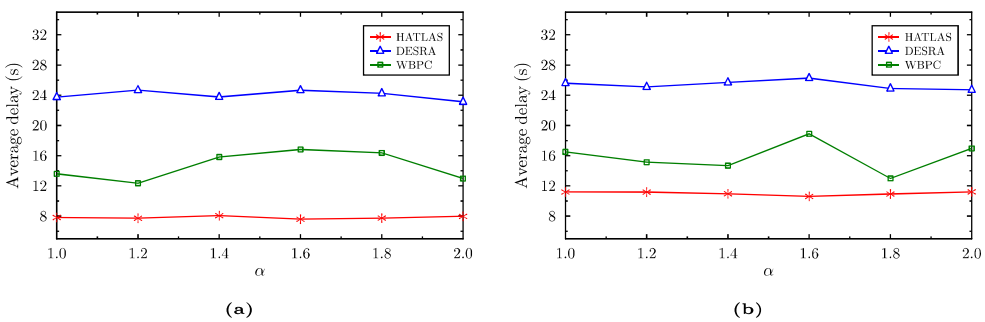


Fig. 6 Average delay at the isolated intersection under: a) homogeneous traffic. b) heterogeneous traffic

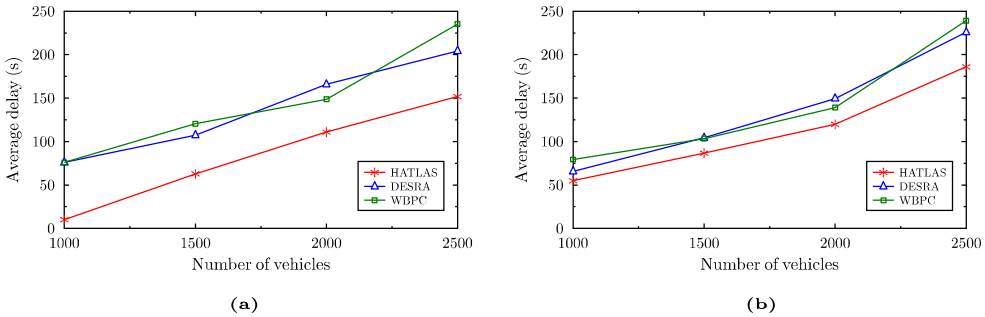


Fig. 7 Average delay in the urban network under: a) homogeneous traffic. b) heterogeneous traffic

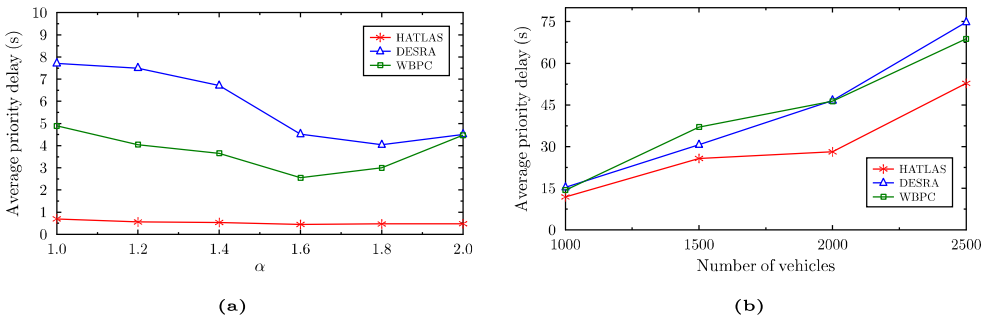


Fig. 8 Average priority delay: a) at the isolated intersection. b) in the urban network

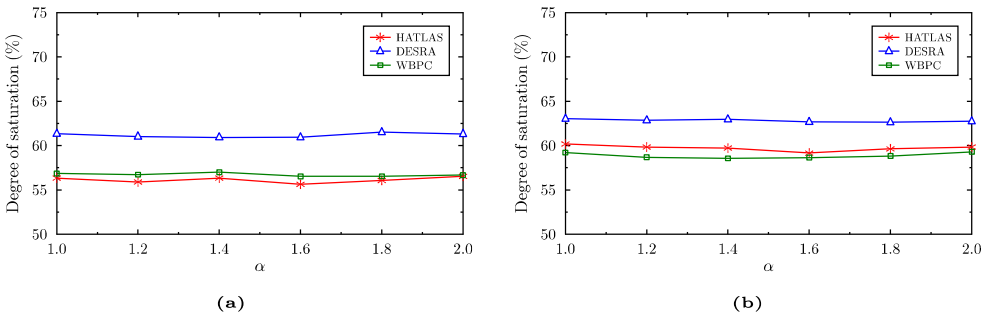


Fig. 9 Degree of saturation at the isolated intersection under : a) homogeneous traffic. b) heterogeneous traffic

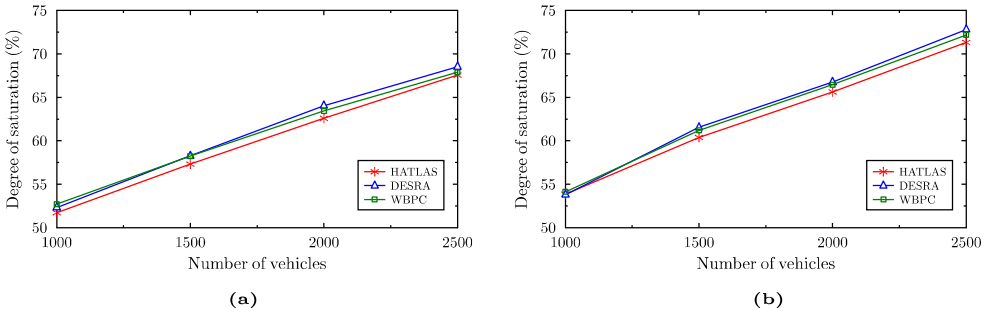


Fig. 10 Degree of saturation in the urban network under : a) homogeneous traffic ; b) heterogeneous traffic

although explicitly controlling spillbacks; this is due to its time-consuming phase improvement process.

As depicted by Fig. 9b under heterogeneous traffic scenarios degrees of HATLAS are slightly higher than those produced by WBPC; this is due to HATLAS' emergency vehicle prioritization scheme that evidently often tends to favour delay over queue length.

In the urban network scenario Fig. 10a shows that under homogeneous traffic, degrees of saturation increase with the number of vehicles. HATLAS yet produces the lowest values due to its downstream edge overflow control process that tends to maximize vehicle throughput during each phase. The second reason is that the critical ratio parameter used to calculate upstream lanes fitness index also helps to control queue lengths and thus to prevent saturation.

Figure 10b suggests that this twofold queue length control scheme helps HATLAS to better prevent lane saturation than WBPC and DESRA; despite a necessarily delay-prone priority management strategy.

5 Conclusion

In this study we addressed the adaptive traffic light control problem aiming to minimise delay and queue length in a realistic urban network. We formulated this question as a variation of the capacitated facility location optimization problem. We designed an integer programming model and an heuristic that leverages both waiting time and queue length to mitigate the spillback phenomenon. The resulting protocol referred to as HATLAS is acyclic, fully distributed and handles both *one-lane-to-one* and *one-lane-to-many* movements. Furthermore, the proposed solution uses a preemption-free strategy to seamlessly cope with emergency vehicle prioritization issues such as deadlock and starvation between emergency vehicles from conflicting queues. HATLAS outperforms two related schemes recently suggested by the literature. These results prove that the queue-and-delay-based schemes implicitly assist in balancing the lack of spillback control; but explicitly applying such a control within this hybrid approach helps to efficiently prioritize vehicles right of ways. Thanks to all these features, HATLAS can be easily used for delay mitigation in real-world situations.

As a future work, we plan on extending this solution to explicitly consider pedestrians, connected autonomous vehicles and bus-stops. We also intend to formally prove HATLAS' ability to reduce gaz emissions then its infrastructure independence through scenarios involving roundabouts.

Declarations

Conflict of Interest The authors declare that they have no conflict of interest.

References

1. Qadri, S.S.S.M., Gökçe, M.A., Öner, E.: State-of-art review of traffic signal control methods: challenges and opportunities. Eur. Transp. Res. Rev. **12**(1). https://doi.org/10.1186/s12544-020-00439-1
2. Eom, M., Kim, B.-I.: The traffic signal control problem for intersections: a review. Eur. Transp. Res. Rev. **12**(1):. https://doi.org/10.1186/s12544-020-00440-8
3. Qiao, Z., Ke, L., Zhang, G., Wang, X.: Adaptive collaborative optimization of traffic network signal timing based on immune-fireworks algorithm and hierarchical strategy. Appl. Intell. **51**(10), 6951–6967 (2021). https://doi.org/10.1007/s10489-021-02256-y
4. Tomar, I., Sreedevi, I., Pandey, N.: State-of-art review of traffic light synchronization for intelligent vehicles: current status, challenges, and emerging trends. Electronics **11**(3), 465 (2022). https://doi.org/10.3390/electronics11030465
5. Agrawal, A., Paulus, R.: Intelligent traffic light design and control in smart cities: a survey on techniques and methodologies. Int. J. Veh. Inf. Commun. Syst. **5**(4), 436 (2020). https://doi.org/10.1504/ijvics.2020.111456
6. Gettman, D., Abbas, M., Liu, H., Skabardonis, A.: Operation of traffic signal systems in oversaturated conditions, volume 2—final report transportation research board. https://doi.org/10.17226/22289 (2014)
7. Mohajerpoor, R., Cai, C.: A traffic signal control strategy to avoid spillback on short links. In: 2020 American Control Conference (ACC), pp. 1167–1172. IEEE (2020). https://doi.org/10.23919/ACC45564.2020.9147494
8. Mohajerpoor, R., Saberi, M., Ramezani, M.: Analytical derivation of the optimal traffic signal timing: minimizing delay variability and spillback probability for undersaturated intersections. Transp. Res. B: Methodol. **119**, 45–68 (2019). https://doi.org/10.1016/j.trb.2018.11.004
9. Noaeen, M., Mohajerpoor, R., Far, B.H., Ramezani, M.: Real-time decentralized traffic signal control for congested urban networks considering queue spillbacks. Transp. Res. Part C Emerg. Technol. **133**, 103407 (2021). https://doi.org/10.1016/j.trc.2021.103407
10. Parihar, A.S., Chakraborty, S.K.: Token-based approach in distributed mutual exclusion algorithms: a review and direction to future research. J. Supercomput. **77**(12), 14305–14355 (2021). https://doi.org/10.1007/s11227-021-03802-8
11. Li, L., Okoth, V., Jabari, S.E.: Backpressure control with estimated queue lengths for urban network traffic. IET Intell. Transp. Syst. **15**(2), 320–330 (2021). https://doi.org/10.1049/itr2.12027
12. Wu, J., Ghosal, D., Zhang, M., Chuah, C.-N.: Delay-based traffic signal control for throughput optimality and fairness at an isolated intersection. IEEE Trans. Veh. Technol. **67**(2), 896–909 (2018). https://doi.org/10.1109/TVT.2017.2760820
13. Li, L., Jabari, S.E.: Position weighted backpressure intersection control for urban networks. Transp. Res. B: Methodol. **128**, 435–461 (2019). https://doi.org/10.1016/j.trb.2019.08.005
14. Younes, M.B., Boukerche, A.: An efficient dynamic traffic light scheduling algorithm considering emergency vehicles for

intelligent transportation systems. Wirel. Netw **24**(7), 2451–2463 (2017). https://doi.org/10.1007/s11276-017-1482-5
15. Gao, K., Zhang, Y., Zhang, Y., Su, R., Suganthan, P.N.: Meta-heuristics for bi-objective urban traffic light scheduling problems. IEEE Trans. Intell. Transp. Syst. **20**(7), 2618–2629 (2019). https://doi.org/10.1109/TITS.2018.2868728
16. Grandinetti, P., de Wit, C.C., Garin, F.: Distributed optimal traffic lights design for large-scale urban networks. IEEE Trans. Control Syst. Technol. **27**(3), 950–963 (2019). https://doi.org/10.1109/TCST.2018.2807792
17. Soon, K.L., Lim, J.M.-Y., Parthiban, R.: Coordinated traffic light control in cooperative green vehicle routing for pheromone-based multi-agent systems. Appl. Soft Comput. **81**, 105486 (2019). https://doi.org/10.1016/j.asoc.2019.105486
18. Zhang, L., Zhao, Q., Wang, L., Zhang, L.: Urban intersection signal control based on time-space resource scheduling. IEEE Access **9**, 49281–49291 (2021). https://doi.org/10.1109/ACCESS.2021.3059496
19. Liu, W.-L., Gong, Y.-J., Chen, W.-N., Zhang, J., Dou, Z.: An agile vehicle-based dynamic user equilibrium scheme for urban traffic signal control. IET Intell. Transp. Syst. **15**(5), 619–634 (2021). https://doi.org/10.1049/itr2.12049
20. Yousef, K.M.A., Shatnawi, A., Latayfeh, M.: Intelligent traffic light scheduling technique using calendar-based history information. Futur. Gener. Comput. Syst. **91**, 124–135 (2019). https://doi.org/10.1016/j.future.2018.08.037
21. Kumar, N., Rahman, S.S., Dhakad, N.: Fuzzy inference enabled deep reinforcement learning-based traffic light control for intelligent transportation system. IEEE Trans. Intell. Transp. Syst. **22**(8), 4919–4928 (2021). https://doi.org/10.1109/TITS.2020.2984033
22. Oza, P., Chantem, T.: A real-time server based approach for safe and timely intersection crossings. In: 2019 IEEE 25th International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA), pp. 1–12. IEEE (2019). https://doi.org/10.1109/RTCSA.2019.8864584
23. Oza, P., Chantem, T., Murray-Tuite, P.: A coordinated spillback-aware traffic optimization and recovery at multiple intersections. In: 2020 IEEE 26th International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA), pp. 1–10. IEEE (2020). https://doi.org/10.1109/RTCSA50079.2020.9203582
24. Zhang, X., Fan, X., Yu, S., Shan, A., Fan, S., Xiao, Y., Dang, F.: Intersection signal timing optimization: a multi-objective evolutionary algorithm. Sustainability **14**(3), 1506 (2022). https://doi.org/10.3390/su14031506
25. Bai, S., Bai, X.: A general framework for intersection traffic control with backpressure routing. IEEE Access **9**, 102125–102136 (2021). https://doi.org/10.1109/access.2021.3096827
26. Kumar, S., Baliyan, A., Tiwari, A., Tripathi, A.K., Jaiswal, B.: Intelligent traffic controller. Int. J. Inf. Technol. https://doi.org/10.1007/s41870-019-00405-8
27. Jian, L.: Multi-objective optimisation of traffic signal control based on particle swarm optimisation. Int. J. Grid Utility Comput. **11**(4), 547 (2020). https://doi.org/10.1504/IJGUC.2020.108464
28. Tunc, I., Soylemez, M.T.: State feedback control for intelligent traffic light systems. In: 2020 28th Mediterranean Conference on Control and Automation (MED). IEEE (2020). https://doi.org/10.1109/med48518.2020.9183224
29. Pirkul, H., Jayaraman, V.: A multi-commodity, multi-plant, capacitated facility location problem: formulation and efficient heuristic solution. Comput. Oper. Res. **25**(10), 869–878 (1998). https://doi.org/10.1016/s0305-0548(97)00096-8
30. Farahani, R.Z., SteadieSeifi, M., Asgari, N.: Multiple criteria facility location problems: a survey. Appl. Math. Model. **34**(7), 1689–1709 (2010). https://doi.org/10.1016/j.apm.2009.10.005
31. Melkote, S., Daskin, M.S.: Capacitated facility location/network design problems. Eur. J. Oper. Res. **129**(3), 481–495 (2001). https://doi.org/10.1016/s0377-2217(99)00464-6
32. Lawler, E.L., Lenstra, J.K., Kan, A.H.G.R.: Generating all maximal independent sets: NP-hardness and polynomial-time algorithms. SIAM J. Comput. **9**(3), 558–565 (1980). https://doi.org/10.1137/0209042
33. Moon, J.W., Moser, L.: On cliques in graphs. Isr. J. Math. **3**(1), 23–28 (1965). https://doi.org/10.1007/bf02760024
34. Mishra, N., Pitt, L.: Generating all maximal independent sets of bounded-degree hypergraphs. In: Proceedings of the Tenth Annual Conference on Computational Learning Theory—COLT '97. ACM Press (1997). https://doi.org/10.1145/267460.267500
35. Eppstein, D.: All maximal independent sets and dynamic dominance for sparse graphs. ACM Trans. Algorithms **5**(4), 1–14 (2009). https://doi.org/10.1145/1597036.1597042
36. Conte, A., Grossi, R., Marino, A., Uno, T., Versari, L.: Listing maximal independent sets with minimal space and bounded delay. In: String Processing and Information Retrieval, pp. 144–160. Springer International Publishing (2017). https://doi.org/10.1007/978-3-319-67428-5_13
37. Fowler, R.J., Paterson, M.S., Tanimoto, S.L.: Optimal packing and covering in the plane are NP-complete. Inf. Process. Lett. **12**(3), 133–137 (1981). https://doi.org/10.1016/0020-0190(81)90111-3
38. Megiddo, N., Tamir, A.: On the complexity of locating linear facilities in the plane. Oper. Res. Lett. **1**(5), 194–197 (1982). https://doi.org/10.1016/0167-6377(82)90039-6
39. Lopez, P.A., Wiessner, E., Behrisch, M.: Microscopic Traffic Simulation Using SUMO. In: 2018 21st International Conference on Intelligent Transportation Systems (ITSC). IEEE (2018). https://doi.org/10.1109/itsc.2018.8569938
40. Omnet++: a public-source, component-based, modular and open-architecture discrete event simulation environment. Official homepage: http://www.omnetpp.org/ accessed on July 2022

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